

Required Datasets for the Updated SMUrF Model:

- **For GPP Calculation ‘main_script_GPP.r’:**
 - Downscaled TROPOMI SIF (needs to be pre-processed):
 - SIF from Caltech (only available up to Jul 2022, - This is what I have used to date): <ftp://fluo.gps.caltech.edu/data/tropomi/>
 - OR SIF from ESA (European Space Agency, I recommend this one going forward): [S5P-PAL Data portal](#)
 - Near Infrared Reflectance Vegetation Index derived from MODIS MCD43A4 BRDF-adjusted reflectance product: [LP DAAC - MCD43A4](#)
 - MODIS land cover data (to identify areas with water): [LP DAAC - MCD12Q1](#)
 - CSIF (to fill in any gaps in TROPOMI SIF data): <https://data.tpdac.ac.cn/en/data/d7cccf31-9bb5-4356-88a7-38c5458f052b/>
 - See Turner et al. 2020 for downscaling method: <https://doi.org/10.5194/bg-17-405-2020>
 - Apply shoreline correction using ‘**Toronto_tree_analysis.r**’ code
 - Requires City of Toronto Forest & Land Cover 2018 dataset: [Forest and Land Cover - City of Toronto Open Data Portal](#)
 - MODIS Land Cover (500m MCD12Q1): [LP DAAC - MCD12Q1](#)
 - Above-Ground Biomass: [GlobBiomass global datasets](#)
 - SIF-GPP relationships for each landcover (should **now** be on GitHub – make sure to use the one converted to TROPOMI SIF wavelengths: ‘GPP-CSIF_Slopes_All_Converted_TROPOMI.txt’)
 - C3:C4 ratio derived from Annual Crop Inventory (ACI) data: [Annual Crop Inventory - Open Government Portal](#)
 - Needs to be pre-processed using ‘**ACI_C3C4_fraction.r**’ code
- **For Reco Calculation ‘main_script_Reco.r’:**
 - Output from GPP calculation
 - MODIS Land Cover (500m MCD12Q1): [LP DAAC - MCD12Q1](#)
 - Air (2m temperature) and soil temperature (Soil temperature level 1) from ERA5: [ERA5-Land hourly data from 1950 to present](#)
 - Impervious surface area (ISA) data (combination of GMIS, ACI, SOLRIS and City of Toronto’s impermeable surface datasets)
 - Need to pre-process, see supplementary material of <https://doi.org/10.1525/elementa.2023.00102> and ‘**Toronto_permeability_plot_GMIS.r**’ see VPRM code directory for this!
 - GMIS data: <https://doi.org/10.7927/H4P55KKF>
 - ACI data: [Annual Crop Inventory - Open Government Portal](#)
 - Southern Ontario Land Resource Information System (SOLRIS): <https://www.arcgis.com/home/item.html?id=0279f65b82314121b5b5ec93d76bc6ba>
 - City of Toronto Impermeable Surface: [Topographic Mapping – Impermeable Surface - City of Toronto Open Data Portal](#)
 - MODIS MOD09A1 reflectance data:
 - Can be downloaded as TIFF files from [AppEEARS](#)
 - Need to specify boundaries of region of interest and year before downloading
 - Need an account for this
 - Must include the red (b01), NIR (b02), and blue (b03) surface reflectance bands plus the quality flag layer (sur_refl_qc_500m), and day of the year

- You must also include the latest file (doy 361) from the previous year and two files from the next year (doy 1 and 9) so that the code can properly interpolate missing EVI data at the start and end of the year.
 - Note in the SMUrF code you need to determine the day of each year which has the minimum EVI at the reference pixel (an arbitrarily chosen forested pixel outside of the city)
- For NEE Calculation (and temporal downscaling of GPP and Reco) ‘main_script_NEE.r’:
 - Output from GPP calculation
 - Output from Reco calculation
 - ERA5 hourly Surface Solar Radiation Downwards (SSRD) and Air Temperature (2m temperature):
[ERA5-Land hourly data from 1950 to present](#)